

PROJECT REPORT

SMART DOOR CAMERA WITH FACIAL RECOGNITION FEATURE FOR THERMAL SCREENING

A project report on an intelligent entry-monitoring system combining face recognition and non-contact temperature screening

Project Domain	Embedded Systems / IoT / Computer Vision
Project Type	Prototype design and system study
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Institution	_____
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Submitted in partial fulfilment of the requirements for the relevant course/project work.

ABSTRACT

The Smart Door Camera with Facial Recognition Feature for Thermal Screening is an intelligent entry-monitoring concept designed to support safer and more convenient access control. A camera captures a visitor's face, and a processing unit detects the face and compares its features with enrolled users. A non-contact infrared temperature sensor measures the person's forehead or skin-surface temperature from a suitable distance. Based on the configured rules, the system can indicate whether the person is recognized and whether the measured temperature is above a selected threshold. It may then operate an electronic lock, display a status message, or send an alert to an administrator.

The project combines embedded hardware, computer vision, sensor interfacing, and decision logic. It can be adapted for homes, laboratories, offices, and controlled entry points. The system is intended as a screening aid—not a medical diagnostic device. Temperature readings are affected by distance, environment, sensor calibration, and the measurement location, so a high reading should be checked using an appropriate validated method. Facial data and temperature records must also be handled with consent and suitable privacy safeguards.

Keywords: Smart door, facial recognition, thermal screening, infrared sensor, access control, embedded system, IoT.

1. INTRODUCTION

Conventional door security systems commonly use keys, PIN codes, RFID cards, or manual checking. These methods can be inconvenient when a card is lost, a code is shared, or a guard must check each visitor. A smart door camera can automate part of the entry process by combining visual identification with a temperature-screening step.

In this project, a camera supplies image data for face detection and recognition. A non-contact infrared temperature sensor provides a surface-temperature estimate. The controller combines these results and gives an output such as “Recognized,” “Not recognized,” “Temperature check required,” or “Access denied.” The exact response depends on the selected safety and access policy.

2. PROBLEM STATEMENT

Manual visitor verification can take time and may produce inconsistent results. Separate camera, temperature-checking, and door-access processes can also be difficult to manage. The proposed system aims to integrate these functions into one prototype while providing clear status indications and maintaining appropriate privacy and safety controls.

3. OBJECTIVES

- To capture a visitor's image using a door-mounted camera.
- To detect a face and compare it with enrolled face data.
- To measure skin-surface temperature without physical contact.
- To combine identity and temperature results using configurable decision logic.
- To provide a clear display, buzzer, indicator, notification, or door-control output.
- To design the system with attention to accuracy, security, privacy, and user consent.

4. SYSTEM OVERVIEW

The system can be divided into six main units: image capture, face processing, temperature sensing, decision logic, user feedback, and door actuation. A microcontroller or single-board computer coordinates the hardware. A processing platform with sufficient computing capability is required for face recognition; the choice depends on

budget, speed, and whether processing is performed locally or on a separate device.

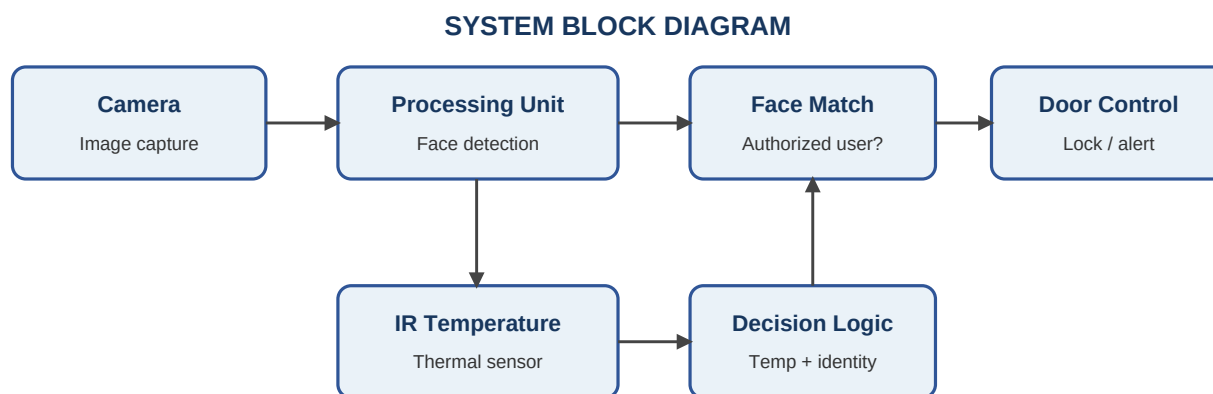


Figure 1. Conceptual block diagram of the proposed smart door screening system.

5. HARDWARE REQUIREMENTS

Component	Purpose	Notes
Camera module / USB camera	Captures the visitor's face image or video.	Use adequate lighting and suitable placement.
Processing unit	Runs face detection, recognition, and decision logic.	Example: Raspberry Pi-class computer or a PC for a prototype.
Non-contact IR temperature sensor	Estimates skin-surface temperature.	Select a sensor designed for the required range and distance.
Microcontroller / GPIO interface	Reads sensor data and controls outputs.	May be integrated into the processing platform.
Display / LEDs / buzzer	Shows identity and screening status.	Provide clear, non-stigmatizing messages.
Relay or electronic lock	Controls a demonstration door lock.	Use safe electrical isolation and a manual override.
Power supply and enclosure	Powers and protects the system.	Follow voltage and current requirements.

6. SOFTWARE REQUIREMENTS

- Operating system and development environment supported by the chosen processing platform.
- Computer-vision library for image capture and face detection.
- Face-recognition method or model, with consent-based enrolment and secure storage.
- Sensor communication library or driver for the infrared temperature sensor.
- GPIO or serial communication software for indicators, buzzer, relay, and display.
- Optional local dashboard or notification service, protected with authentication.

7. WORKING PRINCIPLE

Step 1 – Image capture: When a visitor approaches the door, the camera captures an image or video frame. Good lighting and a clear view of the face improve the quality of the input.

Step 2 – Face detection: The software locates a face in the image. If no face is detected, it requests the visitor to adjust their position and try again.

Step 3 – Facial recognition: The detected face is converted into a feature representation and compared with enrolled users. The software returns a match score. A threshold is selected during testing; the result should not be treated as perfectly accurate.

Step 4 – Temperature measurement: The infrared sensor estimates surface temperature without touching the person. The sensor should be used at its recommended distance and angle, and should be allowed to stabilize according to its datasheet.

Step 5 – Decision: The controller combines the recognition result and temperature reading. For example, access may be permitted only for an authorized person when the screening result is within the configured range. If the reading is high, missing, or unreliable, the system can ask for a repeat measurement or human review.

Step 6 – Output: A display, LED, buzzer, or notification communicates the result. If an electronic lock is included, the prototype controls it according to the chosen policy. A manual exit and override mechanism should always be available.

8. ALGORITHM / FLOW OF OPERATION

1. Start the system and initialize the camera, sensor, display, and door-control interface.
2. Capture an image and check whether a face is detected.
3. If no face is detected, display a positioning message and repeat image capture.
4. If a face is detected, compare it with the enrolled face database.
5. Read the infrared temperature sensor and validate the measurement.
6. Apply the configured rules to identity and temperature results.
7. Display the result, record only necessary event data, and control the lock or alert if enabled.
8. Return to the waiting state for the next visitor.

9. FACIAL RECOGNITION MODULE

Facial recognition typically involves face detection, alignment, feature extraction, and matching. During enrolment, a permitted user's face is captured under suitable conditions and a feature template is created. During operation, a new template is compared with stored templates. The system accepts a match only when the score meets a selected threshold.

Recognition performance may change because of poor lighting, camera angle, face coverings, changes in appearance, or image quality. The prototype should be tested with different users and conditions. For sensitive access decisions, use an additional factor or human confirmation rather than relying solely on facial recognition.

10. THERMAL SCREENING MODULE

A non-contact infrared sensor detects infrared energy emitted by a surface and estimates its temperature. The system uses this estimate as a screening signal. It does not directly measure core body temperature, diagnose infection, or establish whether a person is healthy.

For more consistent readings, the sensor should be installed at the recommended distance, aimed at an appropriate skin area, and used in an environment supported by the manufacturer. Changes in ambient temperature, direct sunlight, sweat, recent outdoor exposure, and sensor calibration can affect the reading. If a reading is above the configured threshold or appears unreliable, request a repeat check using an appropriate validated thermometer and follow local health guidance.

11. DECISION LOGIC

Face result	Temperature result	Suggested system response
Recognized	Within configured range	Show pass status; permit entry if authorized by policy.
Recognized	Above threshold	Do not automatically label the person as ill; request a repeat or human review.
Not recognized	Any result	Deny automatic access or ask the visitor to use the normal verification process.
No face / invalid reading	Unavailable	Ask for repositioning or manual screening; do not make an unsupported decision.

12. APPLICATIONS

- Homes and apartments for convenient door monitoring.
- Office and laboratory entry points.
- Educational institutions and project laboratories.
- Small controlled-access facilities where a prototype can be evaluated.
- Visitor-management systems when combined with clear consent and human oversight.

13. ADVANTAGES

- Combines identity checking and non-contact temperature screening in one system.
- Can reduce repetitive manual steps at selected entry points.
- Provides immediate status through a display or indicator.
- Can be expanded with access logs, alerts, or a local dashboard.
- Supports learning in embedded systems, sensors, computer vision, and IoT.

14. LIMITATIONS

- Face recognition can produce false matches or missed matches.
- Infrared readings are estimates of surface temperature and can be affected by environmental conditions.
- Performance depends on camera quality, lighting, sensor selection, calibration, and processing capability.
- A connected system may face cybersecurity risks if not properly protected.
- The system adds cost and requires maintenance, testing, and user support.

15. PRIVACY, SAFETY, AND SECURITY

Facial images and biometric templates are sensitive information. Obtain informed consent where required, explain the purpose of collection, limit access, and store only the data needed. Prefer local processing when practical, encrypt stored or transmitted information, set a retention period, and provide a way to remove an enrolled user. Do not display a person's identity or temperature result publicly.

A temperature reading should not be used as the sole basis for a medical conclusion. Provide a manual screening option for visitors who cannot or do not wish to use facial recognition. For a real door installation, use certified electrical components, a secure lock interface, emergency egress, and a manual override. Follow applicable privacy, accessibility, workplace, and health requirements.

16. TESTING AND EVALUATION

Test	Method	Expected observation
Face detection	Test under bright, normal, and dim lighting.	Face is detected under supported conditions; unclear cases prompt retry.
Recognition	Test enrolled and non-enrolled participants with consent.	Record correct matches and errors; tune the threshold.
Temperature sensor	Compare repeated readings and a suitable reference thermometer.	Readings are repeatable within the sensor's stated limits.
Decision logic	Try each identity/temperature combination.	System follows the configured decision table.
Fail-safe behavior	Disconnect camera or sensor; simulate invalid readings.	System avoids unsafe automatic approval and provides a clear message.

17. EXPECTED OUTCOME

The expected outcome is a working prototype that captures a face, attempts recognition, obtains a non-contact surface-temperature estimate, and displays a combined status. When connected to a demonstration lock, it can apply a configurable access rule. Actual accuracy and response time must be measured during implementation; they should not be assumed without testing.

18. FUTURE ENHANCEMENTS

- Add a secure administrator dashboard and event notifications.
- Use a liveness or anti-spoofing check to reduce attempts using printed photographs or screens.
- Improve usability with voice prompts and a clearly accessible manual alternative.
- Add local encrypted logs with automatic deletion after a defined period.
- Evaluate recognition accuracy, sensor repeatability, and response time using a documented test plan.

19. CONCLUSION

The Smart Door Camera with Facial Recognition Feature for Thermal Screening integrates camera-based identity checking, non-contact temperature estimation, and entry-control logic. It is a useful educational project for exploring embedded systems, computer vision, sensor interfacing, and IoT. The prototype can improve convenience at selected entry points, but its results must be treated carefully: facial recognition is not infallible and infrared screening is not a medical diagnosis. Proper testing, privacy protection, human review, and safe door-control design are essential before any real-world deployment.

20. REFERENCES

1. Raspberry Pi Documentation, official documentation for hardware and software platform selection.
2. OpenCV Documentation, official documentation for image processing and computer-vision functions.
3. Manufacturer datasheet and application notes for the selected non-contact infrared temperature sensor.
4. Manufacturer documentation for the selected camera, controller, display, relay, and electronic lock.
5. Applicable local privacy, biometric-data, electrical-safety, and public-health guidance.

Note: This is a general project-report draft. Component models, measured results, source code, circuit diagrams, and implementation-specific details should be added after the actual prototype is selected and tested.